SHUTAO ZHENG

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WORK EXPERIENCE

Web Developer | AEC Daily Inc | Newmarket, ON | Co-op Jan. - Apr. 2023 Full-time May 2024 - Present

· Investigated and remediated two blind SQL injection vulnerabilities, eliminating the attack surface by sourcing user identity from session state instead of client input; designed a centralized, boundary-level parameter sanitization layer replacing 4 scattered validation points, with sanitization ordering deliberately sequenced around session and application-stack initialization to prevent unsanitized data propagation.

· Identified and fixed a privilege-escalation vulnerability where staff access was inferred from unverified email domain matching, replacing it with an explicit authorization flag in the user schema.

· Designed an ordered multi-language user preference model, enabling graceful content fallback when courses lacked translations in a user's top-preferred language; extended the gettext-based templating system to render UI text through the same preference chain, unifying content and interface localization under one data-driven mechanism.

· Extended the Live Session Report with multi-session Excel export, and optimized the Link Report to handle large datasets by capping results at 1000 rows with batched cell styling; eliminated a redundant duplicate-processing loop in the Online Event Report to reduce unnecessary computation.

Software Engineer | Lifion by ADP | New York, NY | May - Aug. 2022

· Extended an internal low-code workflow engine used by non-technical staff to define server-side business logic, adding new conditional logic operators to expand the range of business rules users could express without writing code.

· Redesigned the application's output display to surface more detailed execution results, improving visibility into how a configured workflow behaved at runtime.

· Wrote integration tests within the team's existing Jenkins CI pipeline, achieving over 90% coverage of newly introduced functionality to safeguard release stability.

Quality Assurance Engineer | NetDragon Websoft Holdings | Fuzhou, China | Sept. 2020 - Aug. 2021

· Built an automated UI testing system for ActivPanel that simulated user button interactions and validated expected on-screen responses, catching functional defects before release.

· Executed a combination of standardized regression test cases and new feature test cases across three product launches, documenting defects in JIRA and producing release-readiness reports to guide go/no-go decisions.

PROJECTS

CELPIP Exam Practice Platform www.deercelpip.com **JavaScript • Python • PostgreSQL • Git**

· Architected a production English exam practice platform with 30+ Flask REST APIs, powering test delivery, user submissions, AI-based scoring, personalized feedback, and speech-to-text transcription.

· Designed PostgreSQL schemas, indexing, and transaction handling to maintain data consistency under concurrent submissions, supporting thousands of learners with sub-second response times.

· Secured the platform end-to-end with JWT authentication, PBKDF2 password hashing, Fernet encryption, and defenses against SQL injection and XSS.

· Reduced page-load time by 60% through CDN delivery, caching, and compression; validated platform stability under 500+ concurrent users through load testing.

Couples Calendar App **Flutter • Dart • Swift/WidgetKit • CloudBase Serverless**

· Built a local-first Flutter application using Drift (SQLite) with incremental Tencent CloudBase synchronization, implementing last-write-wins conflict resolution to reconcile offline edits across calendar, mood, and journal modules.

· Developed CloudBase serverless functions handling authentication, data synchronization, and partner-interaction features such as habit-tracking check-ins and notifications.

· Created native iOS Home Screen Widgets with Swift and WidgetKit and automated iOS builds and releases through Xcode Cloud.

Personal Portfolio Website stzheng.com **React • JavaScript • Figma • HTML/CSS**

· Designed and built a responsive personal portfolio in React and Figma to showcase professional experience and projects.

EDUCATION

University of Waterloo **Waterloo, Ontario**

Bachelor of Mathematics September 2018 - August 2023

Computational Mathematics Major & Combinatorics and Optimization Minor & Computing Minor

SKILLS

Programming Languages: Python, PHP, JavaScript, Dart, Swift, SQL, C++, HTML/CSS

Frameworks and Libraries: React, Flask, Node.js, Flutter, Smarty, WidgetKit

Development Tools: Git, Figma, Docker, Linux, PostgreSQL, Jenkins, Xcode Cloud, Jira, Bugzilla